

SHANGZHI, China

WMO: 509680

Lat: 45.2061N

Lon: 127.9794E

Elev: 627

StdP: 14.37

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.2	-17.9	-26.8	1.3	-21.8	-22.9	1.6	-17.1	20.2	11.3	17.7	8.4	2.9	220	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	86.8	71.3	84.6	70.6	82.3	70.0	75.8	82.3	74.3	80.6	72.8	78.8	7.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.9	129.4	79.8	72.4	122.6	78.0	70.9	116.4	76.5	40.0	82.6	38.4	80.6	37.1	78.8	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.5	16.1	14.0	-27.8	91.3	6.5	2.5	-32.4	93.1	-36.2	94.6	-39.8	96.1	-44.5	97.9		
			-28.0	78.1	6.4	1.9	-32.6	79.4	-36.3	80.5	-39.9	81.6	-44.5	83.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	39.4	-1.0	7.9	24.8	43.5	56.9	68.2	73.1	70.1	58.7	42.5	22.1	4.0	(6)
		DBStd	27.08	8.21	10.09	10.95	8.08	6.94	5.67	4.06	4.80	6.40	8.03	11.49	9.29	(7)
		HDD50	6312	1581	1179	782	227	16	0	0	9	256	837	1425		(8)
		HDD65	9923	2046	1599	1246	644	265	31	1	13	205	696	1287	1890	(9)
		CDD50	2441	0	0	0	33	230	546	717	622	269	24	0	0	(10)
		CDD65	579	0	0	0	0	14	127	253	170	15	0	0	0	(11)
		CDH74	4195	0	0	0	20	251	1024	1741	1009	147	3	0	0	(12)
		CDH80	1025	0	0	0	1	54	296	463	196	15	0	0	0	(13)
		WSAvg	5.9	5.8	6.1	6.8	7.1	6.7	5.8	5.2	4.9	4.8	5.7	6.0	5.8	(14)
		PrecAvg	26.2	0.2	0.3	0.6	1.3	2.0	4.1	7.0	5.8	2.8	1.3	0.6	0.3	(15)
(16)	Precipitation	PrecMax	42.6	0.4	0.7	2.0	3.3	3.4	10.8	12.2	15.3	6.9	4.3	1.8	0.7	(16)
(17)		PrecMin	17.7	0.0	0.0	0.0	0.3	0.3	0.9	1.8	1.5	0.7	0.2	0.1	0.1	(17)
(18)		PrecStd	5.3	0.1	0.2	0.5	0.9	0.9	2.2	2.8	3.5	1.4	0.9	0.4	0.2	(18)
(19)																
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	28.3	41.0	58.4	77.2	85.1	90.3	90.3	87.2	82.1	72.3	55.7	37.0		(19)
(21)		0.4% MCWB	25.2	32.4	45.6	55.3	62.0	68.5	74.6	74.8	67.7	57.1	46.2	33.1		(20)
(22)		2% DB	21.7	35.8	50.7	70.2	80.8	86.5	87.2	84.6	78.2	66.0	48.6	29.8		(21)
(23)		2% MCWB	18.8	30.2	40.0	51.4	59.6	67.5	73.3	73.5	65.3	53.5	41.6	26.6		(22)
(24)		5% DB	16.9	30.6	45.8	65.1	76.2	83.5	84.9	82.2	74.7	60.8	43.1	23.8		(23)
(25)		5% MCWB	14.6	25.9	36.6	49.3	58.6	66.9	72.9	72.1	62.9	50.5	37.8	21.0		(24)
(26)		10% DB	12.6	25.4	41.3	59.7	71.8	80.2	82.5	79.7	71.5	56.6	38.1	18.8		(25)
(26)		10% MCWB	10.8	21.7	33.8	46.5	56.9	66.3	72.3	70.5	60.7	48.6	33.7	16.6		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	25.6	33.7	46.7	57.2	65.6	73.4	78.6	77.8	70.8	59.1	47.5	33.6		(27)
(28)		0.4% MCDB	28.3	39.1	56.4	70.8	78.4	82.0	86.0	83.9	78.4	69.1	53.9	36.5		(28)
(29)		2% WB	19.0	30.4	41.1	53.1	62.7	71.3	76.5	75.7	68.0	55.6	42.2	26.7		(29)
(30)		2% MCDB	21.6	36.0	49.0	68.0	76.5	80.7	82.8	81.9	74.2	63.0	47.7	29.6		(30)
(31)		5% WB	14.6	26.1	37.4	50.1	60.3	69.6	74.9	74.0	65.4	52.3	38.2	21.2		(31)
(32)		5% MCDB	16.7	30.5	44.6	63.1	71.6	79.1	81.5	79.5	71.8	58.7	42.7	23.8		(32)
(33)		10% WB	10.8	21.8	34.2	47.1	58.2	68.0	73.7	72.4	63.0	49.1	34.1	16.7		(33)
(34)		10% MCDB	12.6	25.4	40.8	58.4	69.2	76.7	80.1	77.4	69.3	55.9	37.7	18.7		(34)
(35)	Mean Daily Temperature Range	MDBR	21.2	24.0	20.6	21.9	21.0	18.1	15.4	15.9	22.0	20.6	17.8	18.2		(35)
(36)		5% DB MCDBR	25.9	27.3	25.0	32.0	29.5	24.0	19.5	19.3	26.4	28.4	22.7	22.8		(36)
(37)		5% DB MCWBR	23.1	21.7	15.5	16.6	12.3	8.7	8.2	9.0	13.4	16.0	16.1	20.0		(37)
(38)		5% WB MCDBR	25.5	26.3	23.5	28.9	24.3	18.8	15.8	15.8	21.2	25.3	21.5	22.7		(38)
(39)		5% WB MCWBR	22.9	21.2	15.3	16.0	11.4	8.0	7.9	8.3	11.7	15.9	16.2	20.2		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.336	0.363	0.441	0.543	0.591	0.610	0.575	0.513	0.458	0.479	0.395	0.350		(40)
(41)		taud	2.025	1.973	1.860	1.730	1.687	1.735	1.850	1.994	2.053	1.882	1.999	2.004		(41)
(42)		Ebn at Noon	229	248	244	229	224	220	226	236	238	204	203	207		(42)
(43)		Edh at Noon	33	43	56	69	75	71	63	52	45	47	35	31		(43)
(44)	All-Sky Solar Radiation	RadAvg	682	1032	1354	1507	1673	1756	1591	1495	1338	933	617	535		(44)
(45)		RadStd	57	69	94	114	162	188	143	119	100	66	49	32		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page